LABORATORY RECORD

Course: Full Stack Web Development
Course Code: 23CM4121
Experiment No.: 3
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1. TITLE

JavaScript ES6 Array Methods and Object-Oriented Programming – Library Management System

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2. AIM

To develop JavaScript programs for array manipulation (calculating total and average marks, adding elements) and implement an Object-Oriented Library Management System using ES6 classes with methods for adding, displaying, totaling, and removing books.

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3. OBJECTIVE

The objectives of this experiment are:

- To write modular JavaScript functions for array data processing.
- To implement total and average calculation algorithms using loops.
- To manipulate arrays dynamically using push() and pop() methods.
- To define ES6 classes with constructors and instance methods.
- To manage collections of object literals inside class instances.

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4. THEORY

JavaScript provides robust array operations and ES6 Classes for Object-Oriented Programming.

Functions encapsulate reusable logic (e.g. iterating over marks to compute totals).

Classes encapsulate state (this.books array) and behaviors (addBook, displayBooks, totalPrice, removeBook), enabling clean, modular software architecture.

Application Flow:
Initialize Array/Class -> Execute Processing Functions -> Manage Book Objects -> Print Output to Console

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5. ALGORITHM / PROCEDURE

1. Create function.js and define an array of student marks [85, 90, 78, 88, 95].
2. Implement displayMarks(), totalMarks(), averageMarks(), and addMark().
3. Calculate and log initial total (436) and average (87.2).
4. Add a new mark (92) and log updated total (528).
5. Create object.js and define ES6 class Library with constructor() initializing an empty books array.
6. Implement addBook(name, price), displayBooks(), totalPrice(), and removeBook().

7. Instantiate Library, add books (Python, Java, HTML, JS), display catalog, calculate total price (■1800), remove a book, and print updated total (■1400).
8. Execute scripts using Node.js/browser console and verify output.

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6. PROGRAM

function.js

```
// Array let marks = [85, 90, 78,
88, 95];

// Function to display marks function
displayMarks(arr) {
console.log("Student Marks:", arr); }

// Function to calculate total
function totalMarks(arr) {
let total = 0;

    for (let i = 0; i < arr.length; i++) {
total += arr[i];     }

    return total;
}

// Function to calculate average function
averageMarks(arr) {     return
totalMarks(arr) / arr.length; }

// Function to add a new mark
function addMark(arr, mark) {
arr.push(mark);     return
arr; }

displayMarks(marks);

console.log("Total Marks:", totalMarks(marks));

console.log("Average Marks:", averageMarks(marks));

console.log("After Adding New Mark:");

addMark(marks, 92);

displayMarks(marks);

console.log("New Total:", totalMarks(marks));
```

object.js

```
console.log("===== Library Management System =====");

class Library {

    constructor() {
this.books = [];     }

    // Add Book
addBook(name, price) {
this.books.push({
title: name,
cost: price         });

        console.log(name + " book added successfully.");
}

    // Display Books
displayBooks() {

        console.log("\nBooks in Library:");

        for (let i = 0; i < this.books.length; i++) {
```

```
            console.log(
(i + 1) + ". " +
this.books[i].title +
" - ■" +
this.books[i].cost             );

        }

    }

    // Calculate Total Price
totalPrice() {

        let total = 0;

        for (let i = 0; i < this.books.length; i++) {

            total += this.books[i].cost;

}

        return total;

}

    // Remove Last Book
removeBook() {

        let removedBook = this.books.pop();

        console.log("\nRemoved Book: " + removedBook.title);

} }

const lib = new Library();

lib.addBook("Python", 500);
lib.addBook("Java", 600); lib.addBook("HTML",
300); lib.addBook("JavaScript", 400);

lib.displayBooks();

console.log("\nTotal Price: ■" + lib.totalPrice());
lib.removeBook();

console.log("\nAfter Removing One Book:");

lib.displayBooks();

console.log("\nNew Total Price: ■" + lib.totalPrice());
```

7. CODE EXPLANATION

Code / Syntax	Explanation
function totalMarks(arr)	Iterates through array elements and computes cumulative sum.
averageMarks(arr)	Divides total marks by array length to compute arithmetic mean.
arr.push(mark)	Appends a new mark to the end of the array.
class Library { constructor() { this.books = []; } }	ES6 class encapsulating the library state with an internal books array.

addBook(name, price)	Pushes an object literal { title, cost } into the books array.
this.books.pop()	Removes and returns the last book from the books collection.
totalPrice()	Iterates through all book objects and calculates total inventory cost.

8. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

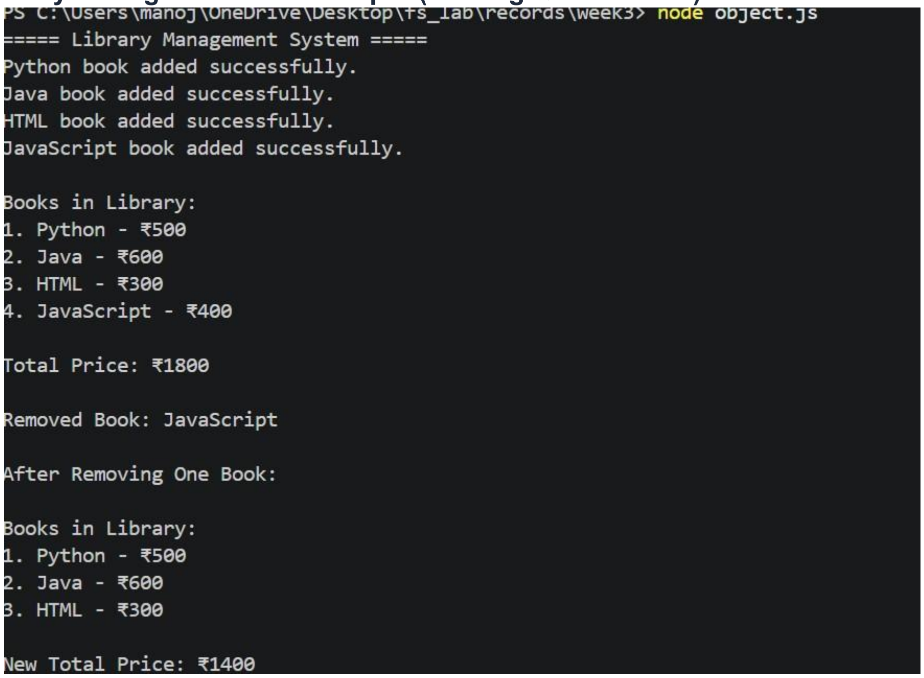
Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Marks Display	Call displayMarks(marks)	Displays initial array [85, 90, 78, 88, 95]	**Pass**
TC-02	Total & Average	Call totalMarks & averageMarks	Total: 436, Average: 87.2 calculated accurately	**Pass**
TC-03	Add Mark	Call addMark(marks, 92)	Mark appended; new total: 528 logged	**Pass**
TC-04	Library Add Books	Add 4 books to Library	Books added and catalog displayed with prices	**Pass**
TC-05	Total Price & Remove	Calculate total and remove book	Initial total ■1800, after removal new total ■1400	**Pass**

9. OUTPUT

Output 1: JavaScript Array Functions Output (Total & Average Marks)

```
PS C:\Users\manoj\OneDrive\Desktop\fs_lab\records\week3> node function.js
Student Marks: [ 85, 90, 78, 88, 95 ]
Total Marks: 436
Average Marks: 87.2
After Adding New Mark:
Student Marks: [ 85, 90, 78, 88, 95, 92 ]
New Total: 528
```

Output 2: ES6 Library Management Class Output (Catalog & Total Price)



```
PS C:\Users\manoj\OneDrive\Desktop\fs_lab\records\week3> node object.js
===== Library Management System =====
Python book added successfully.
Java book added successfully.
HTML book added successfully.
JavaScript book added successfully.

Books in Library:
1. Python - ₹500
2. Java - ₹600
3. HTML - ₹300
4. JavaScript - ₹400

Total Price: ₹1800

Removed Book: JavaScript

After Removing One Book:

Books in Library:
1. Python - ₹500
2. Java - ₹600
3. HTML - ₹300

New Total Price: ₹1400
```

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10. RESULT

Thus, JavaScript array manipulation functions and the ES6 Object-Oriented Library Management System were successfully developed, executed, and verified with accurate outputs.